

SMART HOSTEL MANAGEMENT SYSTEM

Database Systems Engineering and Distributed Backend Development

Course Code: 25CS1302E

Koneru Lakshmaiah Education Foundation

Section No: 13 | Team No: 2

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Abstract

Sec No: 13 | Team No: 2 | Project Title: **Smart Hostel Management System** | Y25 · 2026-27 · Trimester 4

Abstract

The Smart Hostel Management System is a database-based application designed to manage hostel activities in an organized, efficient, and reliable manner. In many hostels, student information, room allocation, fee details, complaints, and other records may be maintained manually. Managing these records manually can result in data duplication, errors, difficulty in finding information, and delays in updating records.

The proposed system provides a centralized platform for storing and managing hostel-related information. It maintains student details, room information, room allocation, fee payment records, complaints, and other important hostel records in a structured database. Authorized users can add, update, delete, and retrieve information whenever required. The system can also help administrators check room availability and maintain accurate records of students and hostel facilities.

The main objective of the project is to simplify hostel administration and reduce manual effort. By using a centralized database, the system improves data consistency, reduces redundancy, and enables faster information retrieval.

Overall, the Smart Hostel Management System aims to provide a simple, secure, and efficient solution for managing day-to-day hostel operations, improving the accuracy of hostel records and saving administrative time.

Problem Statement



Student records may be maintained manually



Room allocation can become difficult to track



Room availability may not be updated quickly



Fee records can become difficult to manage



Complaints may be hard to track systematically



Manual records can lead to **duplication and errors**



Searching and updating may take more time



Lack of centralized information affects efficiency

Traditional management → data errors · delays · reduced administrative efficiency

Motivation



Growing hostel records need **systematic management**



Administrators need **quick access** to student and room data



Centralized storage can improve **consistency**



Digital records can reduce **repetitive manual work**



Structured data supports **faster retrieval**



A **web-based system** can ease routine administration

From manual process to digital solution

Manual Process



Problems



Digital Centralized Solution

Objectives

- 1 Develop a centralized hostel management system
- 2 Maintain student information in a structured database
- 3 Manage hostel rooms and room availability
- 4 Manage student-room allocation
- 5 Maintain fee payment records
- 6 Record and manage complaints
- 7 Provide CRUD operations for authorized users
- 8 Reduce data redundancy and manual effort
- 9 Improve information retrieval
- 10 Provide a foundation for scalable backend development

Literature Survey

Category	General approach	Key feature	Limitation / opportunity
Hostel Management Systems	Software that records residents, rooms and hostel operations	Consolidated hostel records	Scope and integration depth vary; to be reviewed
Database-Based Student Management	Relational databases store and query student profiles	Structured storage, quick search	Linking with room and fee data to be examined
Room Allocation Systems	Assigning students to rooms by rules or manual choice	Occupancy and availability tracking	Handling changes and history to be studied
Web-Based Administrative Systems	Browser-based interfaces to administrative data	Access without local installation	Access control and usability to be evaluated
Database-backed Management Applications	Frontend, backend API and database working together	Separation of concerns, maintainability	API design and scalability to be explored

General descriptions only — no specific papers reviewed yet; no research findings are claimed.

Research Gap

Potential gap areas to investigate



Fragmented hostel
information management



Lack of centralized records



Manual room allocation
processes



Limited integration of room,
fee and complaint
information



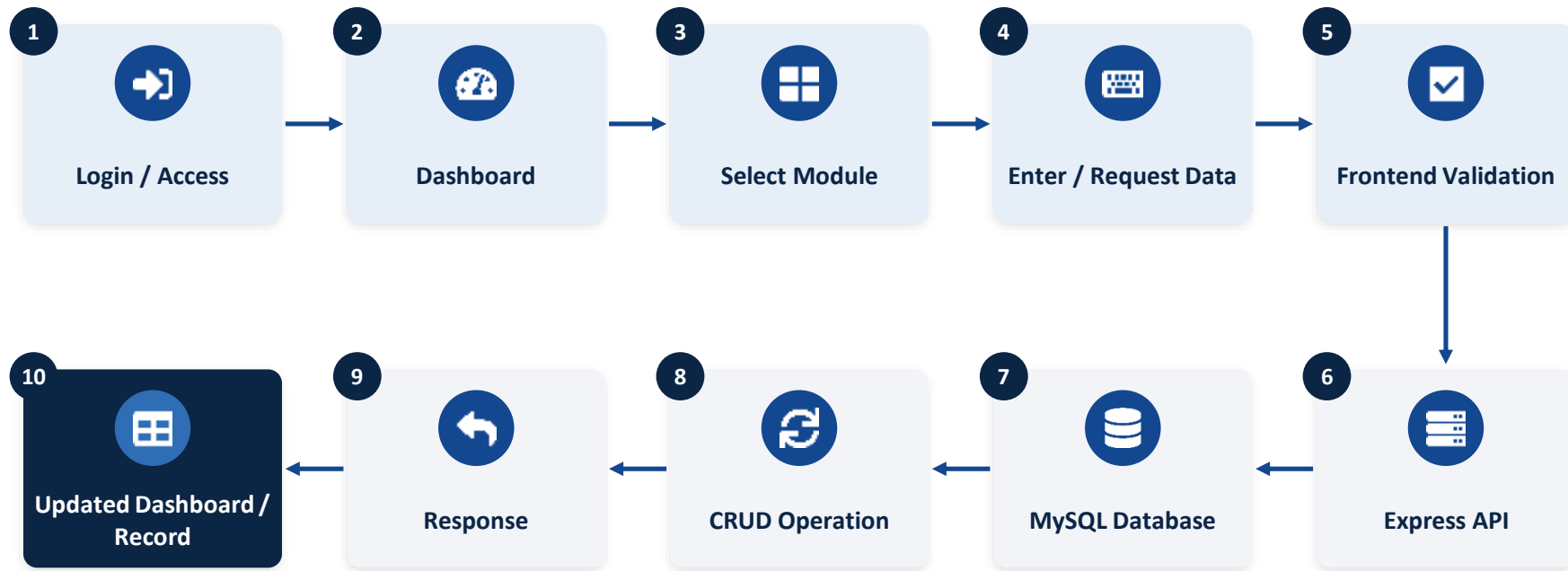
Need for a unified database-
backed web interface



Need for maintainable
backend APIs

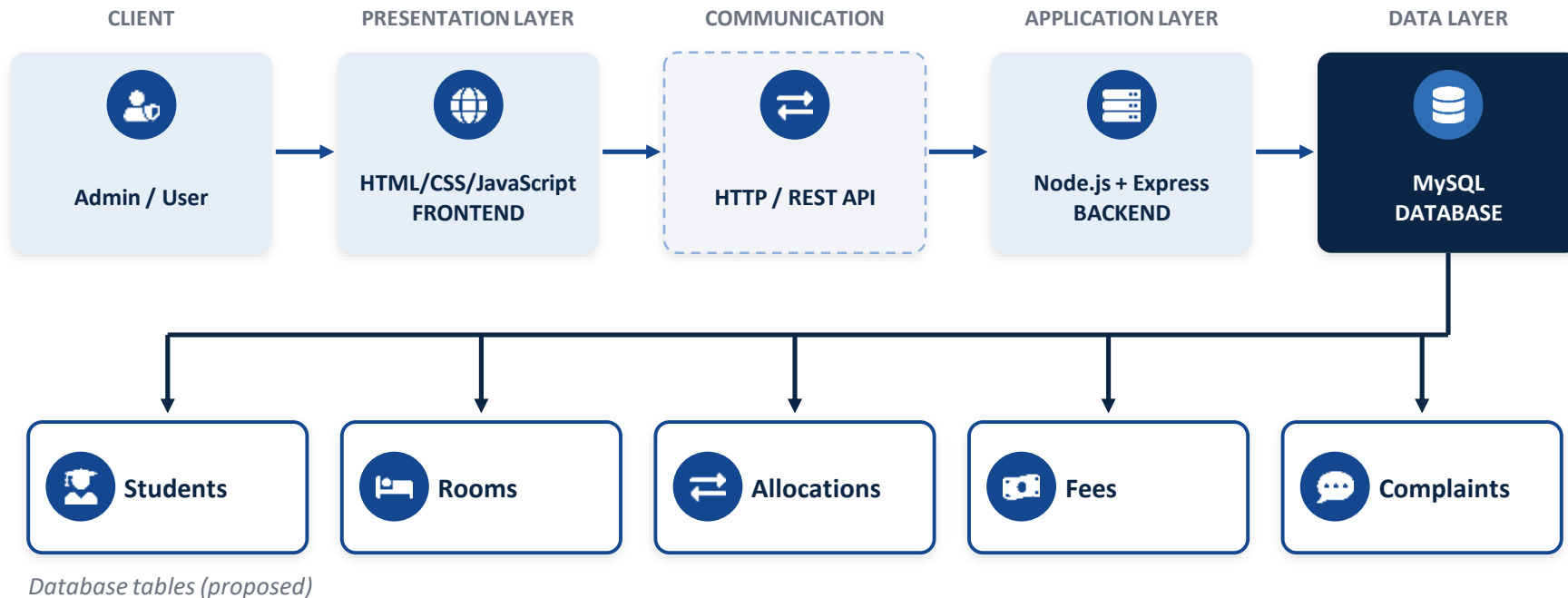
The project will investigate whether a centralized database-backed application can address these management challenges.

Proposed Methodology

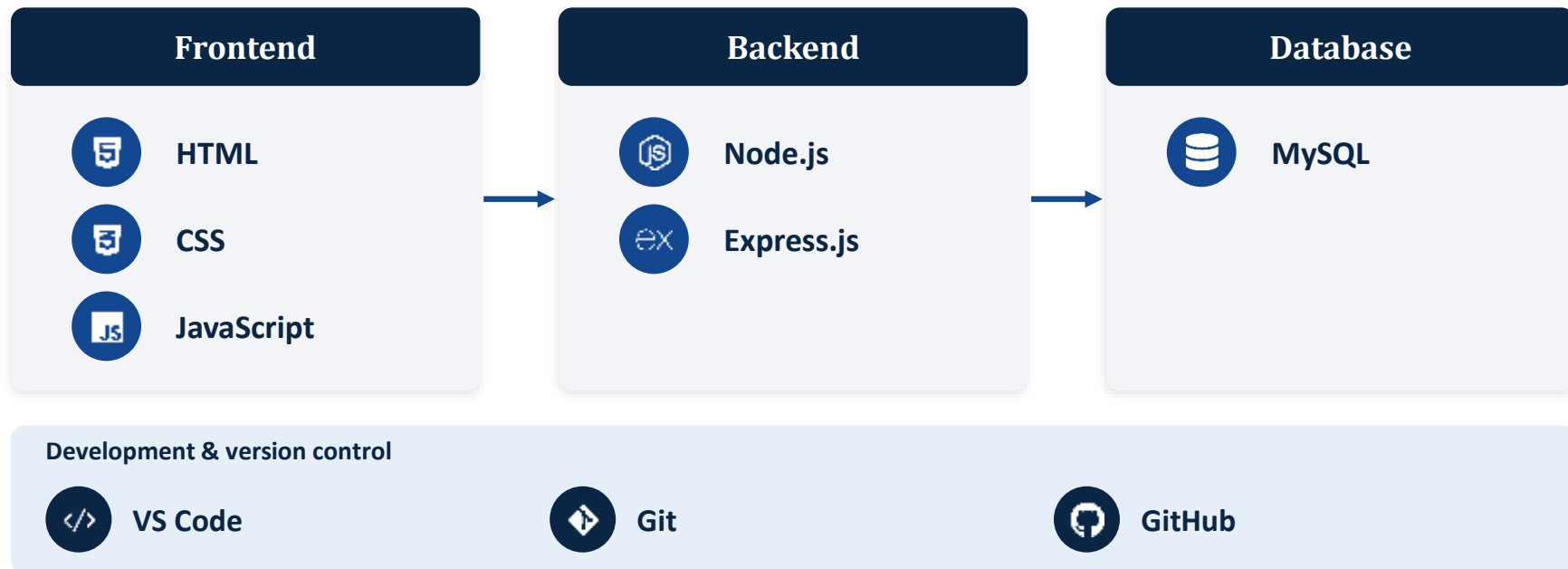


Frontend → Backend → Database → back to the user interface

System Architecture



Technologies Used



These technologies are specified in our project abstract.

Dataset / Input Details

STUDENTS	ROOMS	ALLOCATIONS	FEES	COMPLAINTS
student_id PK	room_id PK	allocation_id PK	fee_id PK	complaint_id PK
name	room_number	student_id FK	student_id FK	student_id FK
email	block	room_id FK	amount	complaint_type
phone	floor	allocation_date	payment_date	description
gender	capacity	status	payment_status	complaint_date
course	occupied_count			status
year	status			

PK Primary key

FK Foreign key

Planned relational tables — the schema has not been created yet.

Implementation – Module 1



1

Student Management

Add, view, update, delete and search student records.



2

Room Management

Maintain room details, occupancy and availability.



3

Room Allocation

Assign or change rooms and keep allocation history.



4

Fee Management

Record payments and view payment status.



5

Complaint Management

Register complaints and update their status.



6

Authentication / Authorization

Login and role-based access to modules.



7

Admin Dashboard

Central place to reach and manage all modules.



8

Reports

Summaries of students, rooms, fees and complaints.

All modules share one MySQL database through the Express REST API.

Implementation – Module 2



Student Management

- Add student
- View student
- Update student
- Delete student
- Search student



Room Management

- Add room
- Update room
- View room
- Track occupancy
- Check availability



Allocation

- Assign room
- Change room
- View allocation history



Fees

- Record payment
- View payment status
- Search payment records



Complaints

- Register complaint
- View complaints
- Update complaint status



Admin

- Manage records
- View reports
- Control access

Implementation – Module 3



Students

GET /api/students
GET /api/students/:id
POST /api/students
PUT /api/students/:id
DELETE /api/students/:id



Rooms

GET /api/rooms
POST /api/rooms
PUT /api/rooms/:id



Allocations

POST /api/allocations
GET /api/allocations
PUT /api/allocations/:id



Fees

POST /api/fees
GET /api/fees



Complaints

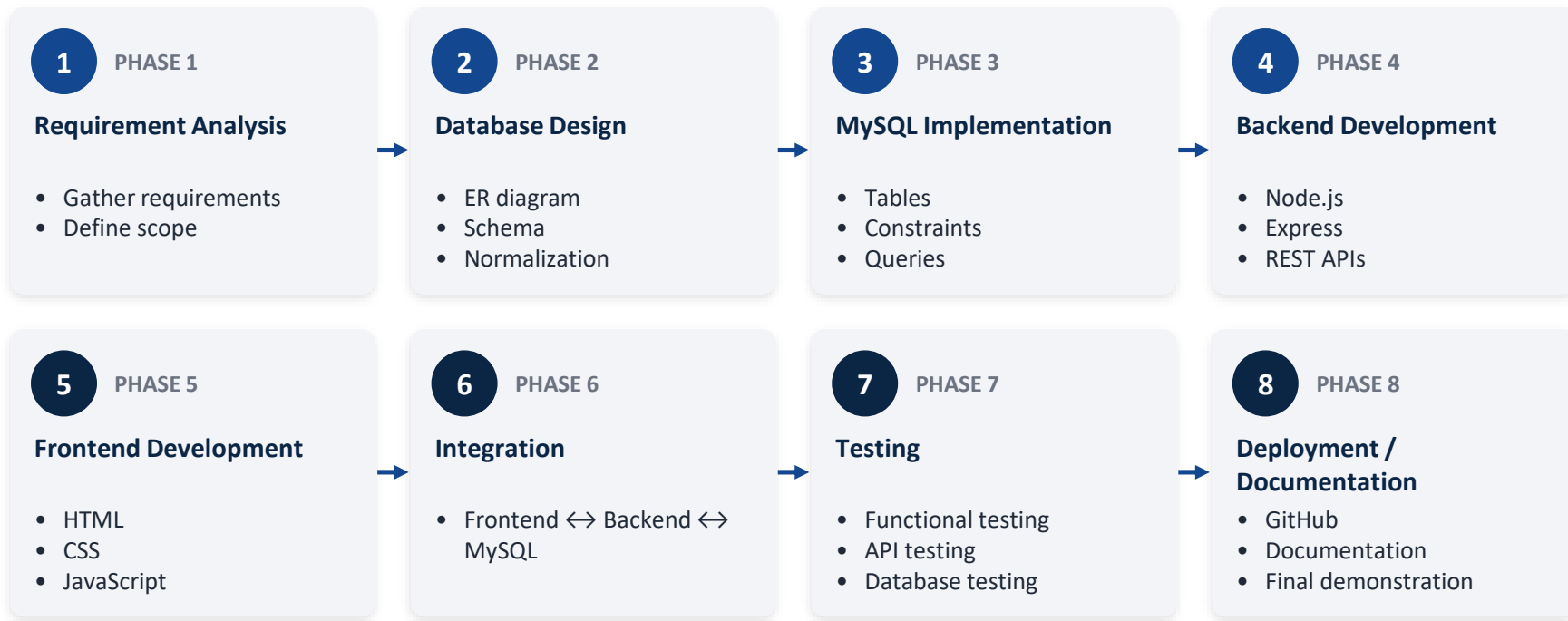
POST /api/complaints
GET /api/complaints
PUT /api/complaints/:id



Design only

These endpoints are proposed. None are implemented or working yet.

Experimental Setup



All phases are planned; implementation has not started.

Results

The expected outcome is a working web-based hostel management system capable of:



Managing student records



Managing room records



Checking room availability



Managing room allocation



Recording fee payments



Managing complaints



Performing CRUD operations



Retrieving information efficiently



Maintaining centralized, structured records

Results will be updated after implementation and testing. No screenshots or test results are shown because none exist yet.

Performance Evaluation



Security

access limited to authorized users



Reliability

consistent, dependable operation



Data consistency

no conflicting copies of data



Usability

simple, clear interface for admins



Performance

fast retrieval and updates



Maintainability

modular, readable code structure



Scalability

able to grow with more records



Availability

accessible whenever needed



Input validation

reject wrong or incomplete data



Error handling

clear messages, graceful failures

Screenshots / Working Demo

Screenshot 1: Admin Dashboard

(to be added after implementation)

Screenshot 2: Room Allocation

(to be added after implementation)

No working demo exists yet. Screenshots and the live demo (GitHub repository) will be added once the modules are implemented and tested.

Comparison with Existing System

Existing / traditional system

-  Paper-based records
-  Spreadsheets / manual files
-  Separate records per hostel activity
-  Manual room allocation
-  Manual fee tracking
-  Manual complaint tracking



Proposed system

-  Centralized MySQL database
-  Structured data, fast retrieval
-  Linked student, room and fee records
-  Assign / change rooms with history
-  Payment records and status
-  Complaints with status updates

Existing approaches are described generally; the proposed features are planned, not yet implemented.

Challenges & Solutions



Data redundancy – fixed by normalization (1NF–3NF)



Linking records – solved with primary and foreign keys



Room overbooking – prevented by capacity and occupancy checks



Unauthorized access – login and role-based access



Wrong or incomplete input – frontend and backend validation



Parameterized queries stop **SQL injection**



Storing passwords – hashing instead of plain text



Frontend–backend integration – REST API design and testing

Anticipated challenges → planned solutions (not yet implemented or tested)

Future Work



Mobile application



Online fee payment integration



Email / SMS notifications



QR-based hostel attendance



Advanced analytics dashboard



Automated reports



Role-based access expansion



Cloud deployment



MongoDB / NoSQL integration where appropriate



Microservice-based architecture



Docker containerization

Possible future extensions — not current features.

Conclusion

The Smart Hostel Management System is proposed as a centralized, database-driven solution for improving hostel administration.

The system aims to:

- ✓ Reduce manual effort
- ✓ Improve data organization
- ✓ Reduce redundancy
- ✓ Improve information retrieval
- ✓ Manage hostel activities systematically
- ✓ Provide a foundation for backend API integration

Next phase

- 1 Database design
- 2 Backend implementation
- 3 Frontend development
- 4 Integration
- 5 Testing

References

1

Course Handout

Database Systems Engineering and Distributed Backend Development (25CS1302E), KL University

2

Project Abstract Submission Form

Smart Hostel Management System, Team 2, Section 13

3

MySQL Documentation

MySQL Reference Manual — dev.mysql.com/doc

4

Node.js Documentation

nodejs.org/docs

5

Express.js Documentation

expressjs.com

6

HTML / CSS / JavaScript Documentation

MDN Web Docs — developer.mozilla.org

7

Git and GitHub Documentation

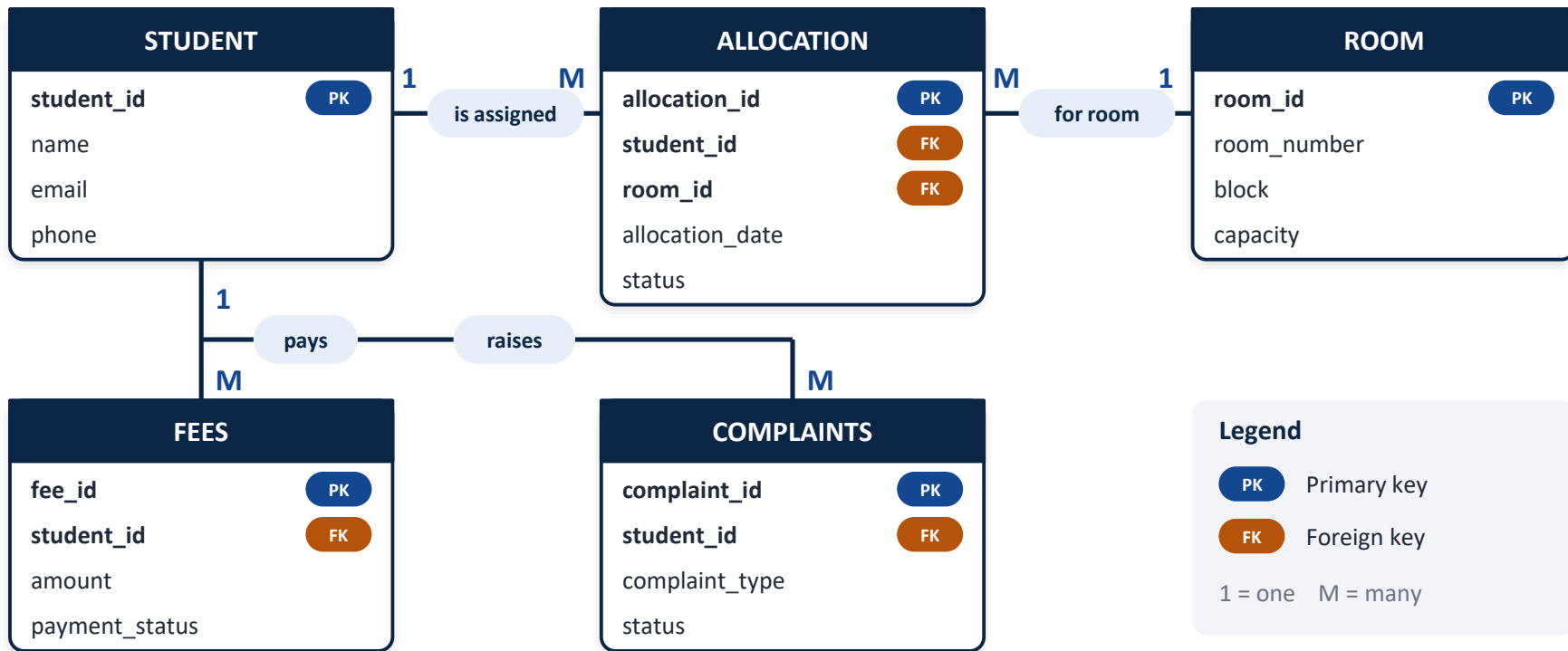
git-scm.com/doc · docs.github.com

8

Academic papers

To be added after an actual literature search

Entity Relationship Diagram



THANK YOU

Questions & Discussion

Smart Hostel Management System

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